CFD Report : Oil Bubble Impact on a Bluff Body in an Upward Water Stream

OpenFOAM v2506, interFoam (VOF) · Case: PhaseInverser/Incompressible · Revised 7 November 2026

This revision keeps the original data and results but corrects inconsistencies, removes claims the data cannot support, and states clearly what is validated and what is not. No new simulation was run. The interface-metrics values were checked against the full chart1_new.csv.

1. Summary

This report documents a three-dimensional, two-phase CFD simulation of an oil bubble carried by an upward water stream into a frustum-shaped bluff body. The aim is to see how the impact deforms and stretches the oil–water interface, as a first step towards crude-oil emulsification and phase-inversion studies.

Main results (single laminar run, 0–1 s, coarse mesh):

- A 6-unit diameter oil bubble starts 11 units below the body in a 30 m/s upward stream. First contact occurs at about t = 0.33–0.35 s, broadly consistent with the simple travel-time estimate of about 0.37 s (11 / 30).
- The interface area stays near its initial value (113.9) until impact, then rises to a peak of 596.4 at t = 0.60 s, **5.24× the initial area**. The impact therefore creates a large amount of new interface, but only transiently.
- After the peak the area falls and ends at 89.1 (0.78× initial) at t = 1.0 s. The animation shows the oil breaking into sheets, perforations, ligaments and small patches, and by t = 1.0 s only two thin ligaments remain.
- **No stable emulsion forms in this run.** The new interface does not persist, and the animation shows few separate droplets.
- **Oil volume is probably not conserved.** The late-time area (89.1) is smaller than that of a sphere holding the full oil volume (113.1), which is impossible if all the oil is still present. Oil loss, most likely numerical, has not yet been quantified, so quantitative results after the peak are unreliable.
- The mesh passes checkMesh, but several metrics are close to their limits (Section 5.1), and the bubble is under-resolved (Section 5.2).

What this report is based on. Files received: the 0/, constant/ and system/ directories, the checkMesh log, the interface-metrics CSV (100 samples, 0.01 s apart) and the post-processing animation (100 frames, oil iso-surface coloured by pressure). Frame n is taken as t = n × 0.01 s, which matches the area curve. The VTK fields were not provided, so the animation and the CSV are the only result data. fvSchemes, fvSolution, time step and Courant number were not provided.

Reliability at a glance

Result	Status
Time of first contact (about 0.35 s)	Reasonably reliable
Area growth to a 5.24× peak at t = 0.60 s	Qualitatively reliable; magnitude mesh- and threshold-dependent
Area decay after t = 0.60 s	Contaminated by probable oil loss
Equivalent d32 and droplet counts	Indicative only, upper bound, valid for t = 0.44–0.60 s at most
Pressure history	Qualitative only (start-up artefact, area-averaging)
Emulsification conclusions	Not supported beyond “transient sheet and ligament break-up”

2. Physical Problem and Geometry

An oil bubble sits in water inside a square duct. Water enters at the bottom and flows upward at 30 m/s. A bluff body is fixed in the middle of the duct above the bubble. Gravity acts in −y, and the oil is lighter than water, so buoyancy also pushes the bubble upward.

The body is bluff on the upstream side (a flat square face) and widens downstream, so the flow separates at the leading edges and the wake sits behind the wider section. The domain volume of 14,365 equals the box volume (14,580) minus the body (215), as expected.

Units. The case files are unit-consistent SI (m, s, kg, Pa) as set up. Lengths, areas and speeds below are therefore in m, m² and m/s unless stated. The fluid viscosities (Section 3) are not physical SI values, so it should be confirmed that no non-dimensional scaling was intended.

Item	Value
Domain (x × y × z)	18 × 45 × 18 (blockMesh scale 1)
Flow direction	+y (inlet at y = 0, outlet at y = 45)
Initial oil bubble	Sphere, radius 3 (diameter 6), centre (9, 6, 9), set with setFields (alpha.water = 0 inside)
Bluff body	Square frustum from the STL file, y = 20 to 25, centred at x = z = 9
Body leading (bottom) face	5 × 5 at y = 20 (x, z from 6.5 to 11.5)
Body top face	8 × 8 at y = 25 (x, z from 5 to 13)
Body volume / surface area	215 / 224.7 (12 triangles, flat-faced geometry)
Blockage	7.7% at the leading face, 19.8% at the widest section (of the 324 cross-section)
Gap, bubble top to body	9 → 20, i.e. 11
Flow-through time of the domain	about 1.5 s (45 / 30)

Note that the bubble (diameter 6) is wider than the leading face (5), so the bubble does not meet the body head-on in a purely centred, small-bubble sense: its outer part is displaced around the face edges, which matters for how the sheet forms.

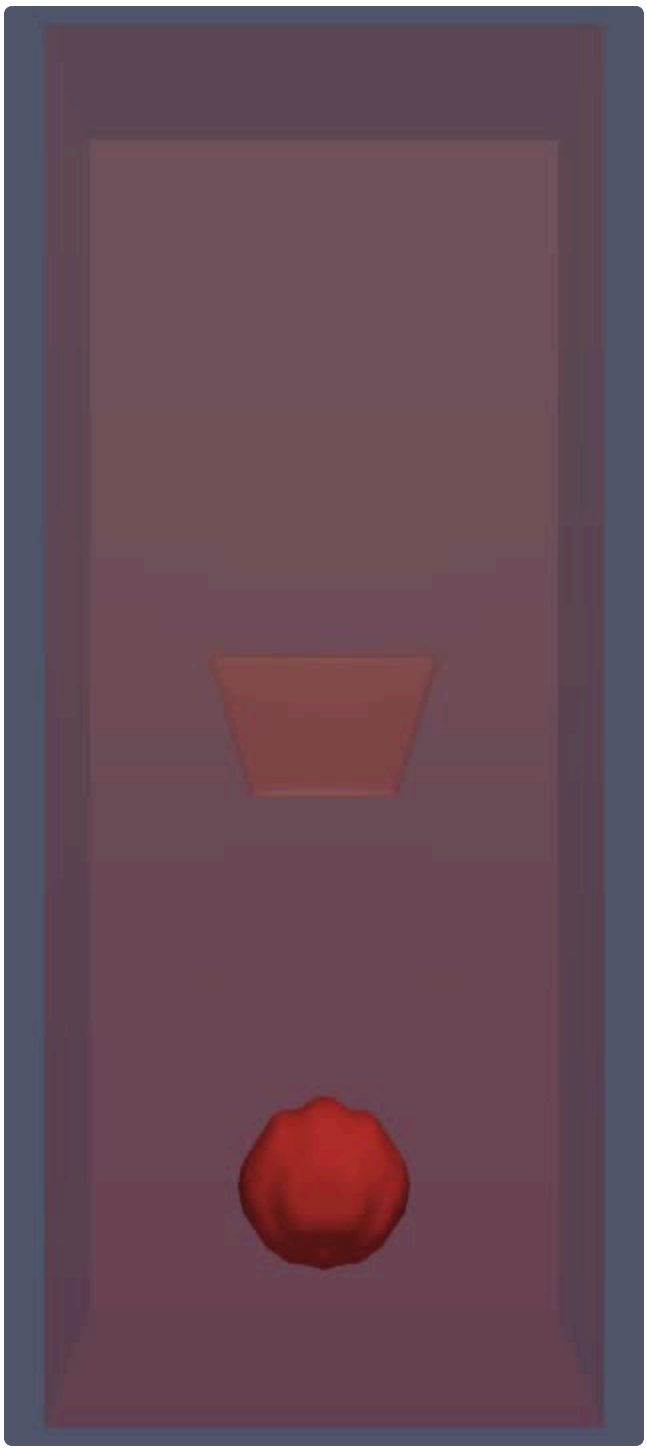


Figure 1. Starting state at t = 0.01 s: the oil bubble (red) sits well below the frustum bluff body, with the water flowing upward (frame 1 of incompressible_new.avi, cropped to the duct).

3. Governing Model and Fluid Properties

The case uses interFoam: incompressible, isothermal, two-phase flow with the volume-of-fluid method and MULES interface compression. The phase list is declared as (water air). **The second phase, named "air", actually represents the oil**, and "nu2" in the CSV is the oil viscosity. Renaming the phase to "oil" is recommended to avoid misreading.

Property	Water (phase 1)	Oil (phase 2, named "air")
Density	1000	800
Viscosity model	Newtonian, $\nu = 1\times10^{-3}$	Cross power-law (shear-thinning)
Cross coefficients	–	$\nu_0 = 1\times10^{-3}$, $\nu\infty = 1\times10^{-5}$, m = 0.4, n = 0.6
Surface tension σ	0.002	
Gravity	(0, −9.81, 0)	
Turbulence	Laminar (no turbulence model)	

Fluid-property caveats

- **Water viscosity.** OpenFOAM's `nu` is kinematic. Real water is about 1×10^{-6} m²/s, so $\nu = 1\times10^{-3}$ is about 1000× too viscous. It equals water's dynamic viscosity in Pa·s, which suggests a units mix-up. If it is deliberate, say so.
- **The oil is not crude-oil-like.** Its viscosity (1×10^{-3} down to 1×10^{-5}) is equal to or below the water value, whereas real crude oil is usually much more viscous. Results should be described as a generic light-oil/water case, not a crude-oil emulsification prediction.
- **Surface tension is low.** σ = 0.002 N/m is roughly an order of magnitude below typical oil–water values (about 0.02–0.03 N/m), so break-up is easier here than in practice.
- **No turbulence model.** At 30 m/s the flow is laminar only by assumption. The finest breakup and wake structure are not turbulence-resolved.

4. Numerical Set-up

4.1 Boundary conditions

All four lateral sides are no-slip walls, so the flow is a confined square duct with developing wall boundary layers, and the inlet profile is flat.

Patch	Velocity U	p_rgh
inlet (bottom)	fixed value (0, 30, 0)	fixedFluxPressure
outlet (top)	pressureInletOutletVelocity	fixed value 0
bluffbody	fixed value 0 (no-slip)	zeroGradient
leftWall, rightWall, front, back	noSlip	fixedFluxPressure

Body-wall pressure condition. The bluffbody uses zeroGradient for p_rgh. With gravity and a no-slip wall, fixedFluxPressure is the usual choice, because zeroGradient can introduce a small spurious flux at the wall. Whether this affects the near-body pressure and the sheet formation should be checked.

Start-up. The domain starts at rest with 30 m/s imposed at the inlet, an impulsive start. The large initial mean p_rgh ($+1.95\times10^{5}$ Pa at t = 0.01 s) is most likely a start-up pressure adjustment, not a physical feature of the bubble approach. The velocity and pressure fields need some time to settle, so early-time values should not be interpreted physically.

Missing set-up information. The time step, maximum Courant and interface Courant numbers, solver tolerances and the fvSchemes (including the exact compression coefficient) were not provided. They should be reported in any publication-grade version.

5. Mesh

The background mesh from blockMesh has 85,284 hexahedral cells in nine blocks. The graded blocks cluster cells toward the side walls. snappyHexMesh then refines around the body (surface level 3–4, feature-edge level 3), snaps to the STL and adds 3 boundary layers on the body (final thickness 0.3 of the local cell, expansion ratio 1.0). The result is 454,578 cells.

5.1 checkMesh results (7 Nov 2026)

The last row of the patch table in the log (a patch named ".*" with 81,124 faces) equals the sum of all boundary faces, so it is a regex-matched summary row and not a real patch.

Quantity	Value	Comment
Points / faces / cells	543,063 / 1,450,564 / 454,578	
Hexahedra	402,586 (88.6%)	
Polyhedra	43,366 (9.5%)	
Prisms / tet wedges	8,518 / 108	
Boundary patches	7 (inlet, outlet, leftWall, rightWall, front, back, bluffbody)	
Bluff-body patch	66,284 faces, closed and singly connected	
Regions	1 (OK)	
Bounding box	(0 0 0) to (18 45 18)	
Min / max cell volume	3.1×10^{-7} / 0.786	Large range
Min / max face area	2.0×10^{-5} / 1.716	
Max aspect ratio	54.5	High
Non-orthogonality	max 65.1°, average 11.8°	Max near the usual limit of about 70°
Max skewness	3.9995	Essentially at the limit of 4
Max cell openness	4.0×10^{-16}	
Verdict	Mesh OK	**Passes, with several borderline metrics**

checkMesh passes, but the maximum skewness is only 0.0005 below the limit, the maximum non-orthogonality is close to the limit and the maximum aspect ratio is high. These worst cells should be located (for example with `checkMesh -writeSets` or a postProcess set) to confirm they are away from the bubble path and the body wake.

5.2 Resolution in the bubble region

The middle block has cells of about 0.30 (x) × 0.46 (y) × 1.0 (z). Where the bubble starts, this gives roughly 20 × 13 × 6 cells across the 6-unit diameter, so the z direction is coarse. The refinement box in the snappy dictionary is defined with zero thickness and does not refine anything, so **the bubble is resolved on the background mesh until it reaches the body**.

Consequences:

- The strongly anisotropic cells (z about 3× larger than x) bias the shape of the deforming bubble.
- Holes, specks and thin ligaments seen in the animation are only a few cells wide, so their size and number are largely set by the mesh and the interface scheme, not by the physics.
- Interface area measured from an iso-surface depends on mesh size and on the contour level, so the 5.24× peak ratio is not mesh-independent until a refinement study shows it is.

The refinement box should be given a finite thickness (or replaced by a refinementRegion covering the bubble path and the body wake).

6. Results: Interface Metrics, 0–1 s

6.1 How the CSV is read

The CSV holds statistics (avg, max, med, min, q1, q3, std, sum) of integrated surface quantities, with one sample (N = 1) per time. The avg, max, med and sum columns are therefore identical. The “Area” of 113.86 at t = 0.01 matches the surface area of the initial sphere ($4\pi \cdot 3^2 = 113.1$) to 0.7%, so the data is read as integrals over the oil–water interface. Other quantities (velocity, pressure, viscosity, alpha) are divided by the area to give area-weighted interface means.

This reading is an inference from the numbers, not something stated in the files, and Sections 6.2–6.4 depend on it. It should be confirmed against the sampling or postProcessing dictionary that generated the CSV. Because the underlying sums are signed integrals, mean velocity components (Ux, Uz) are expected to cancel by symmetry. They are not a measure of lateral motion and are not used below.

6.2 Time history of the interface

Time (s)	Interface area	Area ratio	Mean interface speed	Mean p_rgh (Pa)	Mean interface viscosity (mixture)
0.01	113.9	1.00	32.4	$+1.95\times10^{5}$	3.9×10^{-4}
0.11	113.3	1.00	31.9	$+1.25\times10^{5}$	4.3×10^{-4}
0.21	113.5	1.00	30.8	$+1.3\times10^{5}$	4.5×10^{-4}
0.31	116.5	1.02	27.7	$+1.85\times10^{5}$	4.6×10^{-4}
0.41	174.8	1.54	26.3	$+0.98\times10^{5}$	2.8×10^{-4}
0.51	385.0	3.38	30.0	-1.0×10^{5}	2.2×10^{-4}
0.61	588.7	5.17	28.9	-2.4×10^{5}	1.9×10^{-4}
0.71	366.6	3.22	20.3	-1.9×10^{5}	1.7×10^{-4}
0.81	223.2	1.96	13.5	-0.2×10^{5}	1.9×10^{-4}
0.91	122.4	1.08	12.2	-1.0×10^{5}	2.7×10^{-4}
1.00	89.1	0.78	14.4	-0.55×10^{5}	3.1×10^{-4}

The p_rgh and viscosity columns are rounded to two significant figures from the area-normalised values. Speed is in m/s. Pressure is **p_rgh only**; it is the pressure with the hydrostatic part removed, and must not be compared directly with static pressure p (which differs by ρgh, hundreds of kPa over a 20–40 m height).

“Mean interface viscosity” is the area-weighted viscosity of the **mixture** in interface cells, which blend water and oil, so it is not the oil viscosity itself. Its starting value (3.9×10^{-4}) is already below the oil’s zero-shear value (1×10^{-3}) for this reason.

The area stays flat until about t = 0.35 s, rises steeply to the peak at t = 0.60 s, and falls faster than it rose.

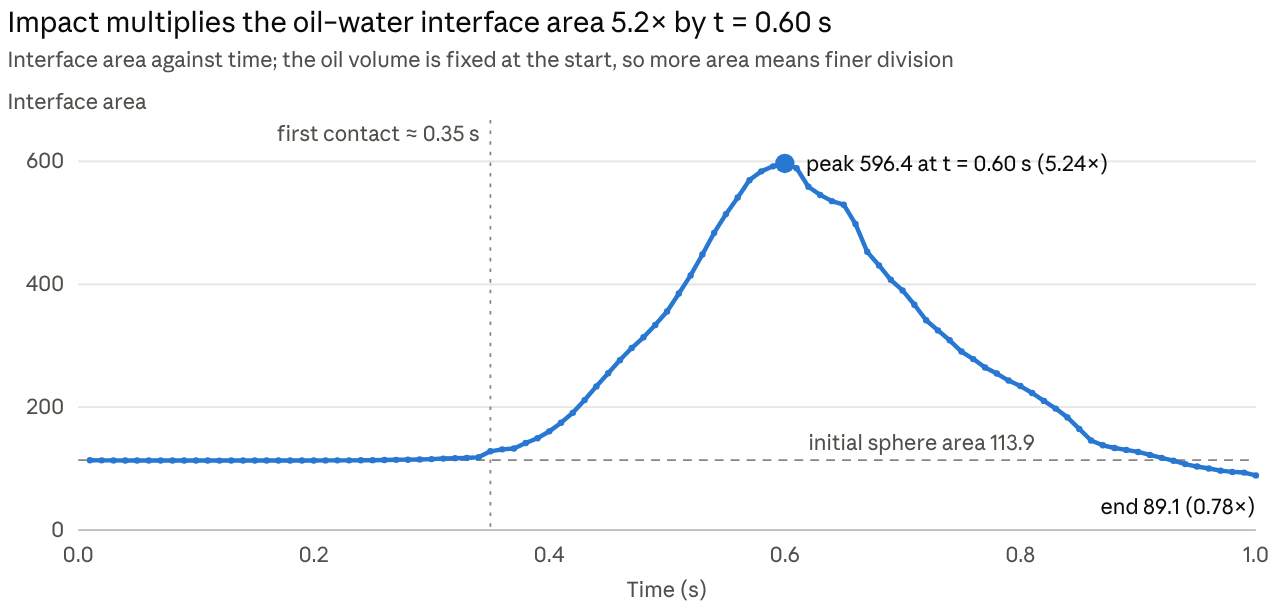


Figure 2. Interface area against time · chart1_new.csv (avg(Area)), all 100 samples, t = 0.01 to 1.00 s

How the curve shows emulsification

Emulsification means one volume of oil is divided into many small pieces. At constant oil volume, division raises the interface area, because a sphere has the smallest area for its volume and every extra fragment, sheet or ligament adds surface. The area ratio therefore measures how finely the oil has been divided.

- **Before impact there is no break-up.** The area stays at about 113.9, the area of the initial sphere, so the bubble keeps its shape on the way to the body.
- **Impact drives the rise.** The area starts to climb at first contact (t ≈ 0.35 s) and reaches 596.4 at t = 0.60 s, 5.24 times the start. This is the same interval in which the animation shows the bubble flatten, wrap the face, perforate and split into ligaments and small patches (Section 6.5). If all the oil were spherical droplets, that area would correspond to about 147 droplets of diameter 1.14, roughly one fifth of the original 6 (Section 6.6).
- **The area stays above twice its initial value for 0.36 s** (t = 0.44 to 0.80 s), so the fine division lasts a measurable time before it decays.

The curve shows strong, impact-driven creation of new interface, which is emulsification-like break-up. It does not show a stable emulsion: the area falls back to its initial value near t = 0.93 s as the sheets thin into a few ligaments, and an area below 113.1 cannot belong to all of the oil, which points to numerical oil loss in this phase. The 147-droplet figure is an upper bound, because sheets and ligaments add area without being separate droplets.

6.3 Flow phases

Phase 1, approach (t = 0 to about 0.30 s). The bubble is carried toward the body. The area is almost constant (minimum 113.3 at t = 0.12), so the bubble keeps a near-spherical shape. The mean interface speed starts at about 32 m/s, a little above the 30 m/s inlet speed (plausibly the impulsive start plus buoyancy; this is an interpretation), and decays to about 28 m/s as the bubble feels the stagnation region ahead of the body. The early pressure values are affected by the start-up artefact (Section 4.1).

Phase 2, impact (about 0.30 to 0.40 s). The area first exceeds its initial value by 5% at t = 0.35 s and doubles by t = 0.44 s. The area-averaged p_rgh reaches a transient maximum of $+2.17\times10^{5}$ Pa at t = 0.36 s. This coincides with impact, but because it is an area average over a deforming interface and the start-up level is already high, it should not be read as the impact stagnation pressure itself. A local maximum of p_rgh on the body face would be the right quantity.

Phase 3, spreading and fragmentation (about 0.40 to 0.60 s). The area grows fastest around t = 0.54 s (about +3,500 per second) and reaches its maximum of **596.4 at t = 0.60 s, 5.24 times the initial area**. The mean p_rgh changes from positive to strongly negative (minimum about -2.7×10^{5} Pa at t = 0.65 s), consistent with fast flow around the body edges and a low-pressure wake. The mean interface viscosity peaks at about 4.6×10^{-4} at t = 0.27 s and falls to a minimum of about 1.7×10^{-4} at t = 0.71 s, which is consistent with shear-thinning responding to strong shear (the Cross law bounds the oil viscosity between 1×10^{-5} and 1×10^{-3}). Both extremes are taken from the full CSV.

Mean pressures of −2 to −5 $\times10^{5}$ Pa (in p_rgh or p, depending on the reference) are large. They should be checked against the vapour pressure for the chosen reference level, because this case includes no cavitation model.

Phase 4, contraction (about 0.60 to 1.00 s). The area falls faster than it rose (about −4,500 per second at t = 0.67 s), returns to the initial value near t = 0.93 s and ends at 78% of it. The mean interface speed drops to about 12–14 m/s, much slower than the 30 m/s stream. The animation shows the face nearly clear of oil by t = 0.90 s and the remaining oil carried upward as ligaments, so the slow mean speed reflects the thin, slow-moving remnant, not oil held on the body. This phase is strongly affected by probable oil loss (Section 6.6).

6.4 Interpretation for emulsification

The peak area ratio (5.2×) is a direct measure of how much new interface the impact creates **at the moment of peak spreading**, so it is the figure to compare across body shapes, inlet speeds and oil viscosities, once its mesh dependence is known. The later fall in area means that area alone does not equal a stable emulsion: within about 0.4 s of the peak the new interface is lost again as the oil thins into a few ligaments (Section 6.5).

A fair indicator of emulsification quality would be the area ratio together with the number and size distribution of oil fragments at a fixed time after impact, and the oil volume. This data set does not contain them.

6.5 Flow visualisation from the animation

The animation shows the oil iso-surface (red) inside the duct, one frame per 0.01 s. Figure 3 picks ten frames (t = 0.01, 0.35, 0.44, 0.54, 0.60, 0.66, 0.70, 0.80, 0.90 and 1.00 s), and the stages below follow them.

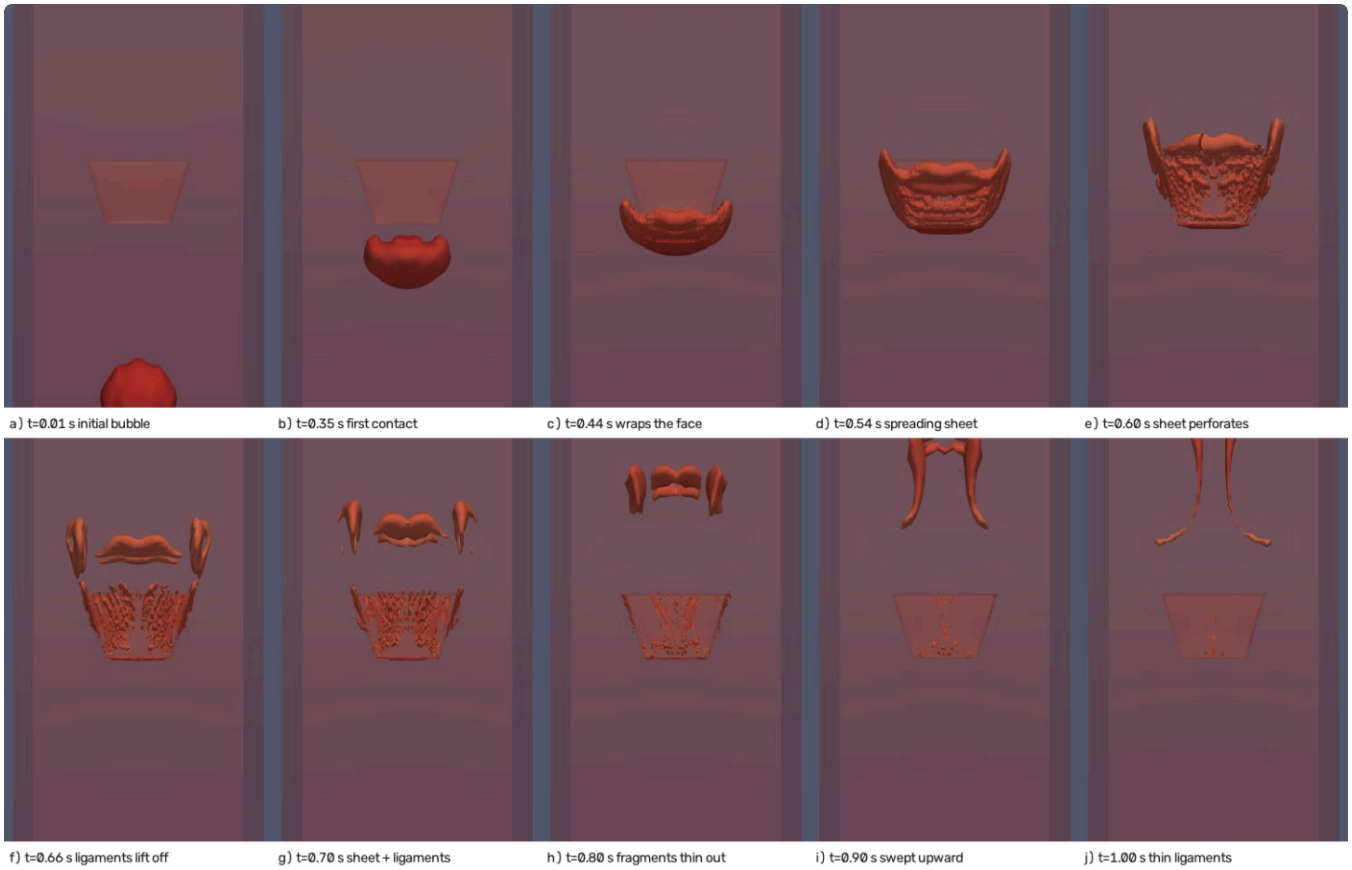


Figure 3. Oil iso-surface at t = 0.01 to 1.00 s, cropped around the bluff body (frames from incompressible_new.avi). The crops leave out the pressure colour bar; the animation's own scale runs from $+5\times10^{5}$ to -6.9×10^{6} Pa.

1. **Approach (t = 0.01 s).** The bubble starts as a slightly irregular sphere well below the body.
2. **Contact (t = 0.35 s).** The bubble has flattened into a dome with two small notches on its rim, and its top is just below the leading face. This is when the area first rises.
3. **Wrapping (t = 0.44 to 0.54 s).** The oil spreads across the 5 × 5 leading face and climbs the sloping sides as a ridged sheet. This is where the area grows fastest.
4. **Perforation (t = 0.60 s).** The sheet develops fine holes and a rough, speckled texture on the face, and two lobes lift off the top corners. The area peaks here.
5. **Break-up (t = 0.66 to 0.70 s).** The oil separates into a central lip-shaped sheet, two side ligaments above the body, and a speckled layer of small oil patches on the face.
6. **Clearing (t = 0.80 to 1.00 s).** The face empties to a few specks, and the oil is carried upward past the body as two thin curved ligaments and a small central piece. At t = 1.0 s only the ligaments remain, still inside the duct and well short of the outlet.

The sequence shows one bubble becoming a sheet, then holes, ligaments and small patches. Two consequences follow:

- Oil does not stay on the body, so wall adhesion does not explain the late area loss.
- The oil visible at t = 1.0 s is far less than the original bubble, and it is still in the duct, so it has not left through the outlet. Oil volume therefore appears not to be conserved.

The finest features (holes and specks) are only a few cells wide, so they are partly set by the mesh. Thin sheets and ligaments of this size are also easily smeared away by the interface scheme, which is a likely cause of the oil loss. This is a hypothesis and has not been tested. **An oil-volume-versus-time check (the volume integral of 1 − alpha.water) is the first thing to do.**

6.6 Evidence of emulsification

Both the interface-area data and the animation show the oil stretched into thin sheets, perforated and broken into ligaments and small patches between t = 0.35 and 0.90 s. This is emulsification-like break-up, but the animation shows few separate droplets, so the equivalent droplet numbers below describe interface size, not a counted dispersion.

For oil of total volume V dispersed as droplets, the interface area A fixes the mean (Sauter) droplet diameter d32 and an equivalent droplet count N_eq. Here V = 113.1, the volume of the initial bubble of diameter 6, which **assumes no oil loss**:

$d32 = 6V / A$, $N_{eq} = V / (\pi\, d32^3 / 6)$

Time (s)	Area ratio (A / initial)	Equivalent d32	Equivalent droplets
0.01	1.00	5.96	1
0.44	2.05	2.90	9
0.50	3.12	1.91	31
0.54	4.25	1.40	78
0.60	5.24	1.14	147

Rows outside t = 0.44–0.60 s are omitted on purpose. At t = 0.35 s the bubble is a deformed dome, so a droplet count of about 1.5 has no meaning. After the peak (for example t = 0.70 s and 0.80 s, with equivalent counts of 41 and 9 in the earlier version) the constant-volume assumption is contradicted by the animation, so those values are not shown.

At the peak (t = 0.60 s) the interface is as large as that of about 147 droplets of diameter 1.14, roughly one fifth of the original 6. The area stays above twice its initial value from t = 0.44 to 0.80 s, a window of 0.36 s.

Limits of this reading

- **The droplet count is an upper bound.** Sheets, ligaments and thin films add area without being separate droplets, so the true number of fragments is lower. The animation confirms that at the peak most of the oil is in sheets and ligaments.
- **Late-time area cannot be read as a droplet size.** A sphere has the smallest area for a given volume (113.1 for V = 113.1), so an area of 89.1 at t = 1.0 s (below 113.1 from t = 0.93 s) cannot belong to all of the oil. Oil is probably being lost numerically.
- **The new interface does not persist.** The sheets thin into a few ligaments rather than a cloud of droplets, so this run does not show a stable emulsion.
- **Numerical, not physical, scales.** With the bubble under-resolved (Section 5.2) and σ and viscosity not representative of crude oil (Section 3), the droplet sizes above are not predictions of a real emulsion.